Knither: Abstracting 2D Knitting

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Introduction

Problem statement: The tooling for producing knitting machine operations is difficult to use. Usually, a user has to write lots of code by hand to process an image and generate Knitout operations. This requires a mastery of the format and very tied to the individual image. We propose to build an image processing pipeline to automate this for 2D swatches with dithering.

Related work:

- The Knitout file format [4] is a platform-independent description of low-level operations shared between different knitting machine models. It's a common format produced by many tools, which can then be compiled to machine-specific operations and validated. It has no control flow or abstraction and is very low level.
- The knitting compiler [3] is the seminal knitting abstraction. It provides an OpenSCAD-like model of composing knit objects out of primitives like tubes and sheets. This enables models to be easily scaled and joints to be rotated. The inputs are expressed graphically.
- KnitScript [2] is a domain-specific programming language that provides facilities for specifying knitting machine operations in a high-level language. They implement control flow and variable bindings. Programs are expressed in terms of primitives: needles, loops, carriers, yarn, sheets, and pass directions. Unlike the knitting compiler, tubes are implemented in terms of sheets. As a limitation, it's more difficult to perform transformations such as flipping or composition.
- Co-Dithering [6] produces double-sided swatches with independent images on either side. Before this work, the back would be the inverse image or a checkerboard. They solve it optimally with dynamic programming.
- Autoknit [5] takes a 3D STL model, identifies whether its 3D surfaces are knitable, and generates Knitout operations to fabricate it. As discussed in the KnitScript paper, this has the limitation that it cannot produce a well-fitted sweater, because 3D models don't encode enough knitting domain information.
- QUILT [1] makes 2D knitting design modular. This enables a user to produce smaller knitting programs that are merged into a larger design.

Approach

Computational: We are building abstractions for the knitting machine. To make it tractable for the class and deliver an interesting result, we are focusing on producing knit swatch patterns from 2D raster images. This will be implemented as an image processing pipeline that produces Knitout operations.

The knitting machine has a limited color palette, allowing up to 4 colors in a horizontal span or up 10 total colors in an object. This makes dithering a good fit for representing images with more colors. We will use several objects as case studies as we increase the capabilities of our system: for two-color dithering, we will use the Utah teapot swatch; for four-color, Alexis' sunset swatch; and for adaptively switching colors to handle images with more than four colors, Alexis' Pokémon pixel art idea. We will pull in ideas from the Co-Dithering paper.

As a stretch goal, once we master producing Knitout instructions for 2D objects, we could branch out into building an OpenSCAD-style object composition system. We would have sheet and tube primitives like the knitting compiler and since sheets and tubes are rectangular, we could apply our

2D raster image texturing to 3D objects. If we were to apply textures to scanned models, we would need a surface-stable dithering algorithm like *Obra Dinn* instead.

Fabrication: We are using the Shima Seiki industrial knitting machine. It will be challenging to learn the knitting terminology and idioms. Since there's limited access to the machine and DAT to 000 compiler, the feedback loop for the fabricability of our Knitout output will be difficult. We have the Knitout to DAT compiler, which will catch some errors, and plan on lots of iteration and collaboration in the knitting lab.

Resources and budget: We will be using existing resources established in the knitting lab. We do not expect to incur much of a budget burden.

Expected Results

Final demo: We aim to have a completed pipeline for processing a raster image to a knit swatch using four colors. As a stretch goal, we could develop an algorithm to utilize colors more than four colors adaptively across an image. We will show swatches produced from a variety of images illustrating challenges in the system.

Midterm milestone: By this milestone, we aim to complete bitonal image processing and fabricate a teapot or similar image.

Bibliography

[1] Jack Hester, Sebastian Law, and Megan Hofmann. 2025. QUILT: Supporting Modular Design of Machine-Knitting Programs. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25)*, 2025. Association for Computing Machinery, Busan, Republic of Korea. https://doi.org/10.1145/3746059.3747608

[2] Megan Hofmann, Lea Albaugh, Tongyan Wang, Jennifer Mankoff, and Scott E Hudson. 2023. KnitScript: A Domain-Specific Scripting Language for Advanced Machine Knitting. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*, 2023. Association for Computing Machinery, San Francisco, CA, USA. https://doi.org/10.1145/3586183.3606789

[3] James McCann, Lea Albaugh, Vidya Narayanan, April Grow, Wojciech Matusik, Jennifer Mankoff, and Jessica K. Hodgins. 2016. A Compiler for 3D Machine Knitting. *Proceedings of ACM Transactions on Graphics (TOG)* 35, 4 (July 2016).

[4] James McCann. The Knitout (.k) File Format. Retrieved October 23, 2026 from https://textiles-lab.github.io/knitout/knitout.html

[5] Vidya Narayanan, Lea Albaugh, Jessica Hodgins, Stelian Coros, and James McCann. 2018. Automatic Machine Knitting of 3D Meshes. *ACM Trans. Graph.* 37, 3 (August 2018), 35:1–35:15. https://doi.org/10.1145/3186265

[6] Yue Xu and James McCann. 2025. Co-Dithering for Jacquard Knitting. In *Proceedings of the ACM Symposium on Computational Fabrication (SCF '25)*, 2025. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3745778.3766646